

Modelling and Rendering of an Apple USBC Power Adapter Using the RenderMan Python API

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Object Analysis and Modelling

The selected object was an Apple USB-C power adapter. Reference photographs were taken from multiple viewpoints to analyse its overall shape and surface characteristics (Figure 1). The charger consists of a large rounded outer body connected to a smaller plug section containing two cylindrical metal pins. The front face contains a recessed grey inset surrounded by a thin white border, while the plug section includes printed regulatory information and symbols. Although the object has a relatively simple overall form, careful observation was required to capture the proportions of the different components and the transitions between the curved and flat surfaces.

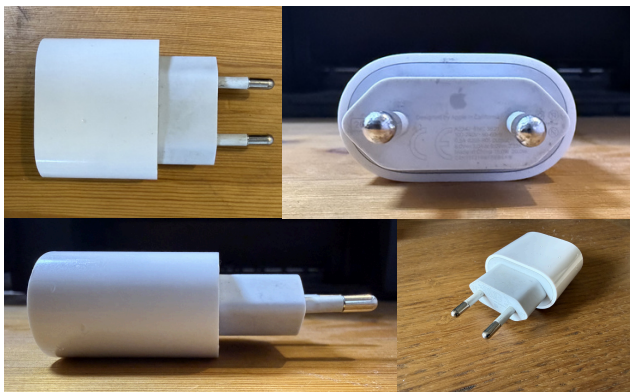


Figure 1: Reference photographs of the Apple USB-C power adapter used during modelling and material analysis.

The charger geometry was created directly in Python using the RenderMan API. The main body was generated from a custom outline describing the cross-sectional shape of the charger, which was then converted into polygonal geometry using PointsPolygons. Additional geometry was created for the front inset, plug section, printed label surface and metal pins. During development, the proportions and outline points were adjusted repeatedly to better match the reference photographs. This approach provided direct control over the shape of the object while keeping the implementation entirely within the RenderMan Python workflow.

Materials and BRDF Selection

The charger contains several visually distinct materials despite being a relatively simple object. The main outer body is manufactured from glossy white plastic, while the plug section and recessed front inset use a darker and less reflective grey plastic. The electrical pins consist of a grey plastic base with metallic tips that exhibit stronger reflections than the surrounding surfaces. In addition, the charger contains printed information and symbols on the front face. Reproducing these material differences was important for making the rendered object resemble the reference photographs.

All materials were implemented using the PxrDisney BRDF. This shader was initially selected because it provides a wide range of controls within a single material model and was capable of reproducing all of the materials required for the project. Different appearances were achieved by adjusting parameters such as base colour, metallic response, roughness and specular intensity. The glossy white body used a relatively low roughness and increased clearcoat to reproduce the reflective plastic surface, while the grey inset used a higher roughness value to create a more matte appearance. The metal pins tips were assigned a fully metallic material with stronger reflections, and separate materials were created for the printed label and wooden table surface.



Figure 2: Different material types visible on the charger model, including glossy plastic, matte plastic, printed label and metal surfaces.

Textures and Surface Detail

The charger contains printed regulatory information and symbols on its front face, while the surrounding scene includes a wooden table surface. These details contribute to the overall realism of the rendered image and help distinguish different surfaces within the scene.

The printed information on the charger was reproduced using an image texture mapped onto the label surface. A separate wood texture was used for the table material. To introduce additional surface detail, a custom OSL shader was developed to generate wood grain patterns, which were applied through a PxrBump node. During look development, the wood texture was colour corrected to create a warmer appearance that better complemented the final lighting setup.

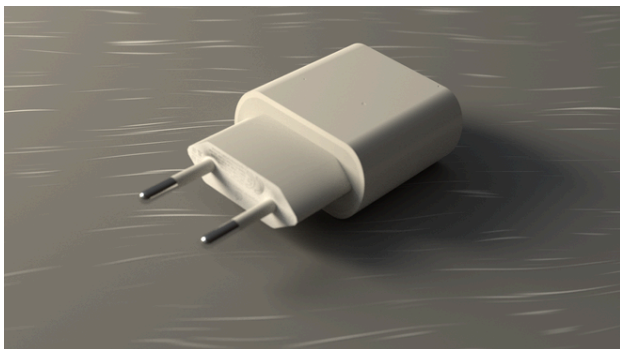


Figure 3: Output of the custom OSL wood shader used to generate additional wood grain detail for the table surface.

Procedural Surface Variation

Although the charger has a relatively clean appearance, the reference photographs show small imperfections and dents on the plastic surface caused by regular use. These subtle details help prevent the object from appearing overly perfect and contribute to a more believable final render.

To reproduce these imperfections, a custom OSL shader was developed to generate a procedural dent pattern across the plastic body. The shader used noise-based controls to determine the placement, size and shape of the dents while avoiding a uniform distribution. Additional parameters were introduced to control edge softness, depth and overall density. The shader was compiled into a .oso shader object using the oslc compiler before being loaded into the RenderMan shading network. The generated pattern was then used to influence

multiple material properties, including colour, roughness and bump information.

As a result, the dents not only altered the shape of the surface but also changed the way light interacted with the plastic, making the wear more visible under different lighting conditions.

Several iterations of the shader were tested during development by adjusting parameters controlling dent density, size and edge softness. This allowed the appearance of the wear pattern to be refined until it produced subtle imperfections without becoming visually distracting.



Figure 4: Close-up view of the procedural dent shader showing variation in dent size, shape and distribution.

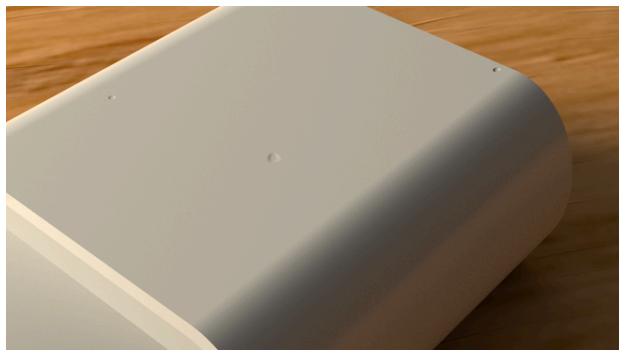


Figure 5: Close-up render showing the dents on the charger.

Lighting, Scene Setup and Camera Effects

The final scene was designed to remain relatively simple so that the focus remained on the charger model. A wooden table was used as the ground surface, providing context without introducing unnecessary complexity. The lighting approach was inspired by product photography, where a small number of carefully positioned light sources are often used to emphasise material properties and surface detail.

The scene was lit using a HDRI environment map from Poly Haven in combination with a rectangular area light. The HDRI provided overall

environmental lighting and reflections, while the area light was positioned to create a stronger directional light source and more defined shading across the charger. The final images were rendered using the PxrPathTracer integrator. Depth of field was applied to both camera views to introduce a photographic effect and help direct attention towards the charger while softly blurring parts of the surrounding scene.

Final Results

Figures 6 and 7 show the completed charger model rendered using the final materials, textures, lighting setup and camera configuration. The first image was designed to closely match one of the reference photographs and focuses on accurately reproducing the proportions, materials and overall appearance of the original object. The second image presents the charger from an alternative viewpoint, allowing additional details such as the surface materials, lighting response and depth of field effect to be observed.

The final renders reproduce the main visual characteristics of the reference object, including the glossy plastic housing, matte grey components, metallic pins and printed label. The use of procedural surface variation helped break up the uniform appearance of the plastic material, while the wood texture and custom bump shader provided additional detail within the surrounding environment. The combination of HDRI lighting, a rectangular area light and depth of field contributed to a more photographic presentation of the final scene.

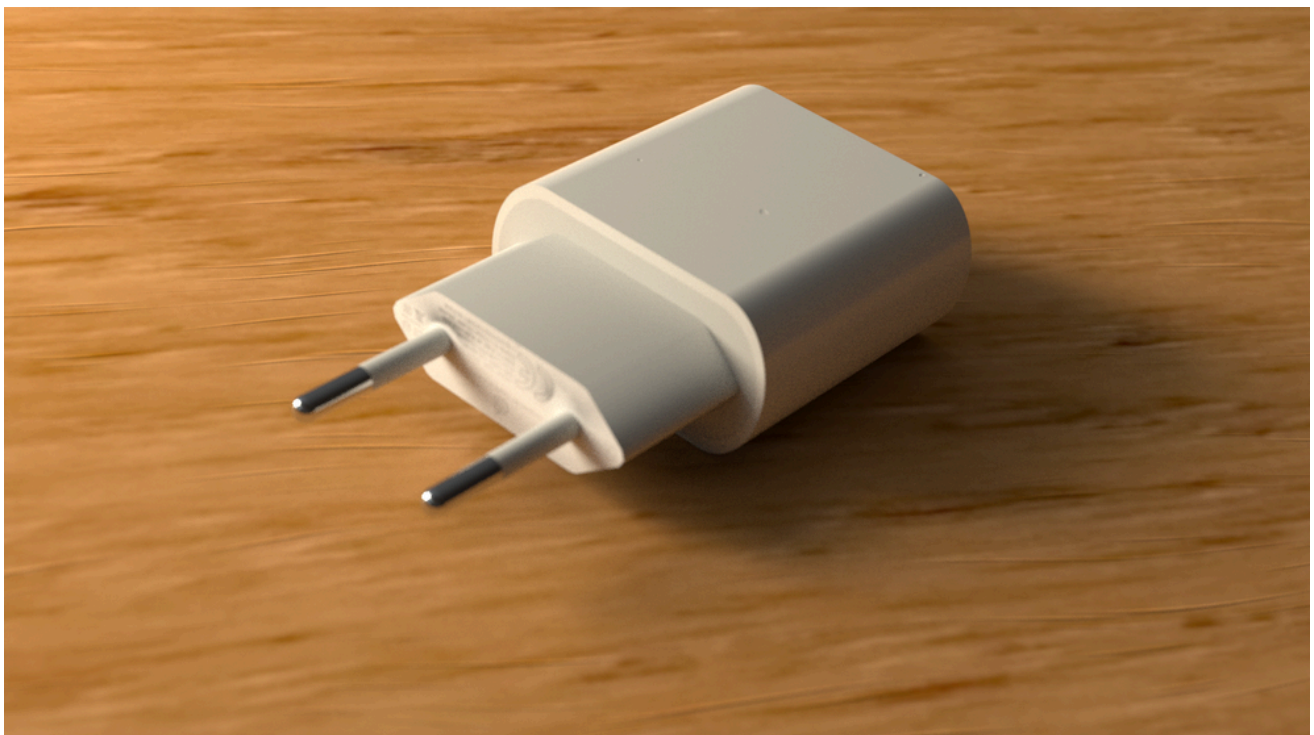


Figure 6: Final render from the primary camera angle.



Figure 7: Final render from an alternative camera angle.

Conclusion

This project demonstrated the complete process of recreating a real-world object using the RenderMan Python API. Reference photographs were used to guide the modelling, material development and lighting decisions throughout the project. The final scene combined custom geometry, physically based materials, image textures, procedural OSL shaders and photographic camera effects to produce two rendered images of the charger. Overall, the project provided practical experience with the different stages of a production rendering workflow while demonstrating the capabilities of RenderMan for modelling, shading and image generation.

References

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